



Trabalho 03 - Linux, Shell Script e Automação Operacional aplicada à Cloud

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Container sendo executado -

The screenshot shows the Docker Desktop interface. On the left, a sidebar contains navigation options: Gordon (BETA), Containers (selected), Images, Logs, Volumes, Kubernetes, Builds, Models, Docker Hub, Docker Scout, MCP Toolkit (BETA), and Extensions. The main area displays 'Containers' with a search bar and a toggle for 'Only show running containers'. Below this, a table lists the running containers:

	Name	Container ID	Image	Port(s)	CPU (%)	Memory usage	Memory (%)	Disk read/write	Network I/O	PIDS	Actions
☒	trabalho03-cloud-shell	-	-	-	0.01%	5.92MB / 7.31GB	0.08%	0B / 12.3KB	1.7KB / 126B	56	[Stop] [Refresh] [Logs] [Delete]

Below the table, a terminal window shows the command prompt and the output of the 'docker ps' command:

```
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
ed78f70bba30   trabalho03-cloud-shell-forum-linux  "bash -c 'service ap..."  9 minutes ago Up 9 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   trabalho03-linux
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell>
```

Volume Docker configurado -

The screenshot shows a terminal window with the command 'docker inspect trabalho03-linux' and its output. The output is a JSON object providing detailed information about the container:

```
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell> docker inspect trabalho03-linux
[
  {
    "Id": "ed78f70bba30e3dbaf088a8a1ad83e16ae4f8cf9517578d881cf3a35e0c331e24",
    "Created": "2026-06-07T11:48:47.380208614Z",
    "Path": "bash",
    "Args": [
      "-c",
      "service apache2 start \u0026\u0026 tail -f /dev/null"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 4119,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-06-07T11:48:47.77594712Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:a1018796087f5e814c6a694f6aa806aa13f04f356e9dd2cd83f1429a17bb4f8",
    "ResolvConfPath": "/var/lib/docker/containers/ed78f70bba30e3dbaf088a8a1ad83e16ae4f8cf9517578d881cf3a35e0c331e24/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/ed78f70bba30e3dbaf088a8a1ad83e16ae4f8cf9517578d881cf3a35e0c331e24/hostname",
    "HostsPath": "/var/lib/docker/containers/ed78f70bba30e3dbaf088a8a1ad83e16ae4f8cf9517578d881cf3a35e0c331e24/hosts",
    "LogPath": "/var/lib/docker/containers/ed78f70bba30e3dbaf088a8a1ad83e16ae4f8cf9517578d881cf3a35e0c331e24/ed78f70bba30e3dbaf088a8a1ad83e16ae4f8cf9517578d881cf3a35e0c331e24-json.log",
    "Name": "/trabalho03-linux",
    "RestartCount": 0,
    "Driver": "overlayfs",
    "Platform": "linux",
    "MountLabel": ""
  }
]
```

Scripts com permissão de execução -

```
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell> docker exec -it trabalho03-linux bash
root@ed78f70bba30:/app#
root@ed78f70bba30:/app#
root@ed78f70bba30:/app# ls -la /app/scripts/
total 52
drwxr-xr-x 1 root root 4096 Jun  7 11:40 .
drwxr-xr-x 1 root root 4096 Jun  7 11:48 ..
-rwxr-xr-x 1 root root  848 Jun  7 11:38 01_update.sh
-rwxr-xr-x 1 root root  815 Jun  7 11:38 02_apache.sh
-rwxr-xr-x 1 root root 1024 Jun  7 11:39 03_estrutura.sh
-rwxr-xr-x 1 root root 1005 Jun  7 11:39 04_backup.sh
-rwxr-xr-x 1 root root  948 Jun  7 11:39 05_deploy.sh
-rwxr-xr-x 1 root root  993 Jun  7 11:39 06_processos.sh
-rwxr-xr-x 1 root root   0 Jun  7 11:39 07_monitoramento.sh
-rwxr-xr-x 1 root root 1234 Jun  7 11:40 08_usuarios_permissoes.sh
-rwxr-xr-x 1 root root 1607 Jun  7 11:40 09_relatorio.sh
-rwxr-xr-x 1 root root 1798 Jun  7 11:40 menu.sh
root@ed78f70bba30:/app#
```

Execução do script de atualização -

```
root@ed78f70bba30:/app# bash /app/scripts/01_update.sh
[2026-06-07 12:00:43] Iniciando atualização do sistema...
[OK] apt update concluído.
[OK] Pacotes atualizados com sucesso.
[2026-06-07 12:00:48] Atualização finalizada.
root@ed78f70bba30:/app#
```

Instalação e validação do Apache -

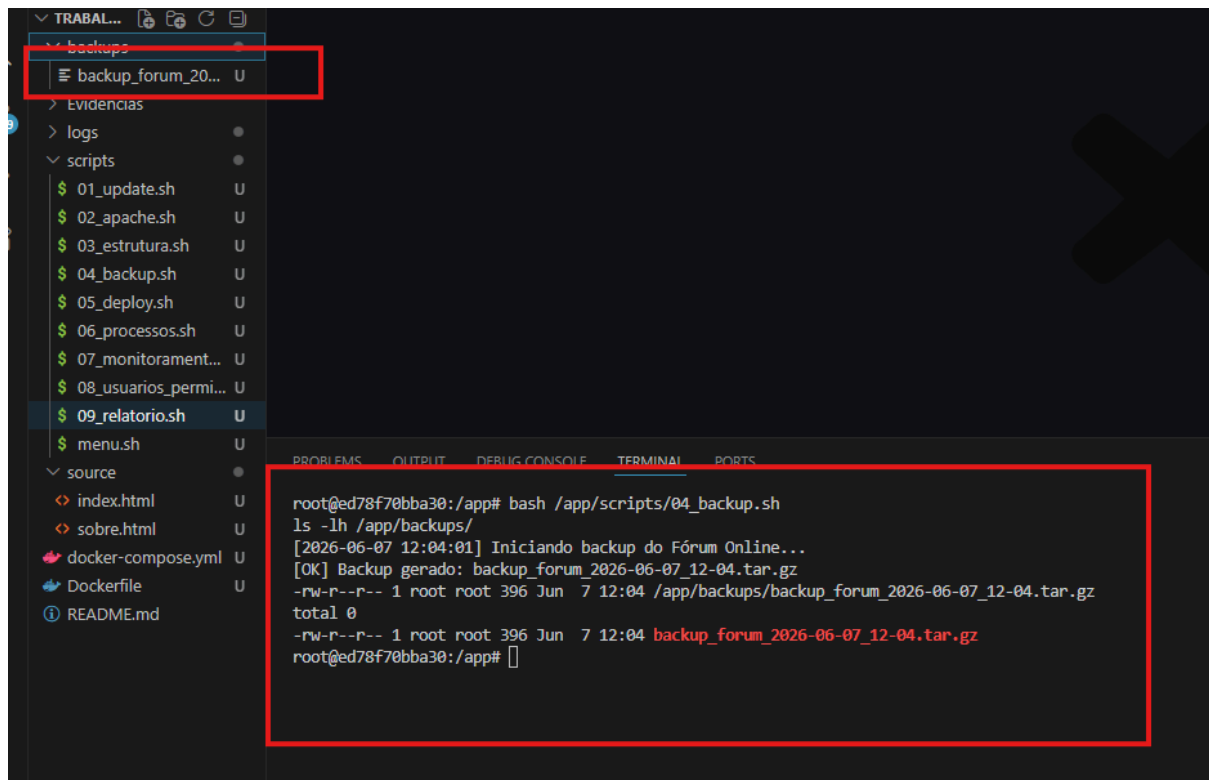
```
root@ed78f70bba30:/app# bash /app/scripts/02_apache.sh
[2026-06-07 12:01:24] Instalando Apache...
[OK] Apache instalado.
[OK] Apache está em execução.
[INFO] Versão instalada: Server version: Apache/2.4.52 (Ubuntu)
root@ed78f70bba30:/app#
```

Estrutura de diretórios criada -

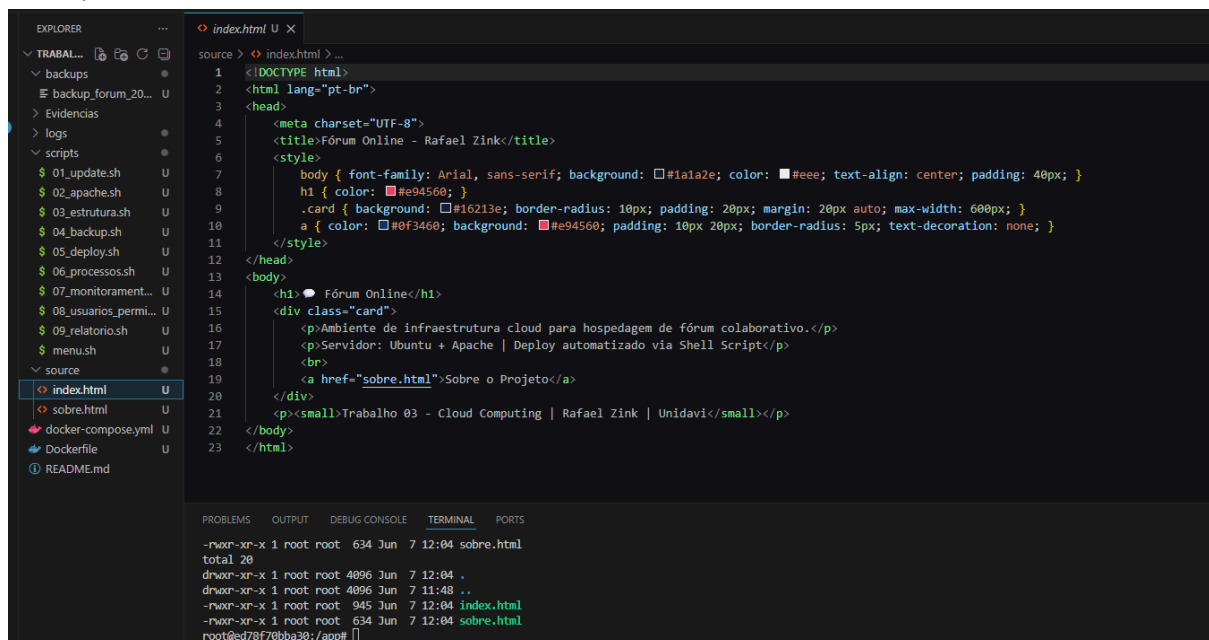
```
EXPLORER
...
✓ TRABALHO03-CLOUD-SHELL
  > backups
  > Evidencias
  > logs
  > scripts
    $ 01_update.sh U
    $ 02_apache.sh U
    $ 03_estrutura.sh U
    $ 04_backup.sh U
    $ 05_deploy.sh U
    $ 06_processos.sh U
    $ 07_monitorament... U
    $ 08_usuarios_permi... U
    $ 09_relatorio.sh U
    $ menu.sh U
  > source
    <> index.html U
    <> sobre.html U
  <> docker-compose.yml U
  <> Dockerfile U
  <i> README.md

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
root@ed78f70bba30:/app# bash /app/scripts/03_estrutura.sh
ls -la /app/forum/
[2026-06-07 12:01:47] Criando estrutura do Fórum Online...
[OK] Estrutura criada com sucesso:
total 44
drwxr-xr-x 10 root root 4096 Jun  7 12:01 .
drwxr-xr-x  1 root root 4096 Jun  7 12:01 ..
drwxr-xr-x  2 root root 4096 Jun  7 12:01 anexos
drwxr-xr-x  2 root root 4096 Jun  7 12:01 backups
drwxr-xr-x  2 root root 4096 Jun  7 12:01 categorias
drwxr-xr-x  2 root root 4096 Jun  7 12:01 logs
drwxr-xr-x  2 root root 4096 Jun  7 12:01 moderacao
drwxr-xr-x  2 root root 4096 Jun  7 12:01 publicacao
drwxr-xr-x  2 root root 4096 Jun  7 12:01 topicos
drwxr-xr-x  2 root root 4096 Jun  7 12:01 usuarios
total 44
drwxr-xr-x 10 root root 4096 Jun  7 12:01 .
drwxr-xr-x  1 root root 4096 Jun  7 12:01 ..
drwxr-xr-x  2 root root 4096 Jun  7 12:01 anexos
drwxr-xr-x  2 root root 4096 Jun  7 12:01 backups
drwxr-xr-x  2 root root 4096 Jun  7 12:01 categorias
drwxr-xr-x  2 root root 4096 Jun  7 12:01 logs
drwxr-xr-x  2 root root 4096 Jun  7 12:01 moderacao
drwxr-xr-x  2 root root 4096 Jun  7 12:01 publicacao
drwxr-xr-x  2 root root 4096 Jun  7 12:01 topicos
drwxr-xr-x  2 root root 4096 Jun  7 12:01 usuarios
root@ed78f70bba30:/app#
```

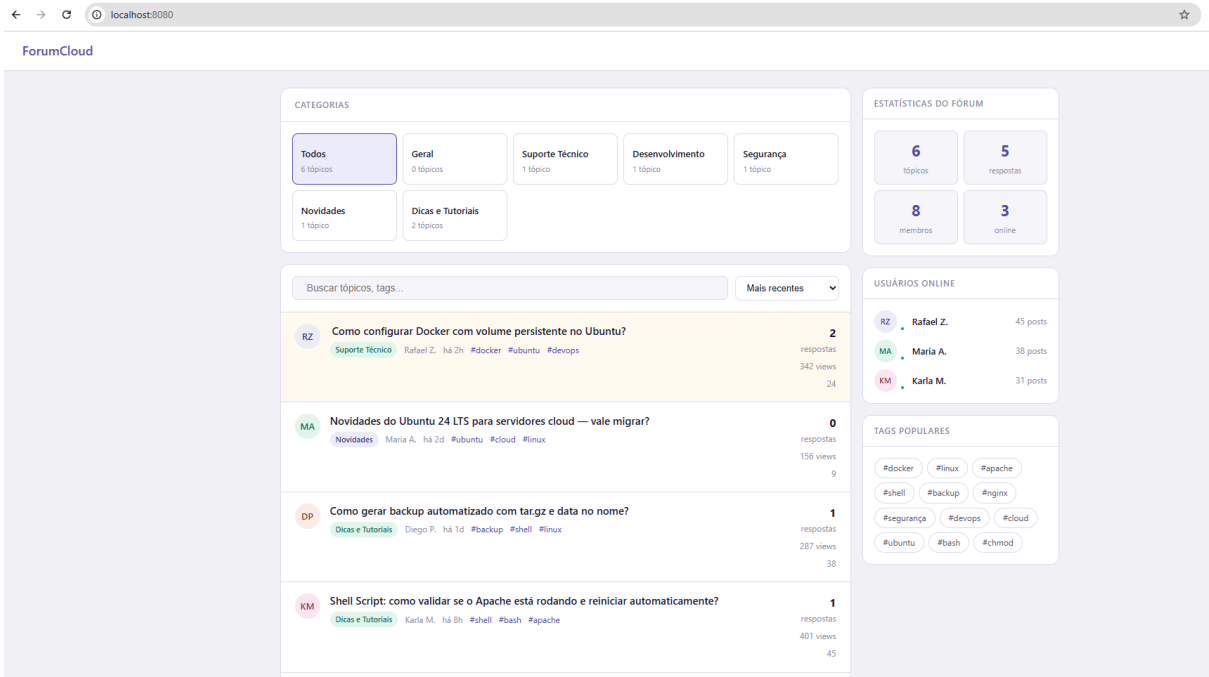
Backup .tar.gz gerado -



Deploy realizado para /var/www/html



Site acessível no navegador -



Usuários e permissões configurados -

```
root@ed78f70bba30:/app# bash /app/scripts/08_usuarios_permissoes.sh
grep "forum_ops\|moderador_user" /etc/group /etc/passwd
ls -la /app/forum/
[2026-06-07 12:06:53] Configurando usuários do Fórum Online...
[OK] Grupo 'forum_ops' criado.
[OK] Usuário 'moderador_user' criado.
[OK] Permissões aplicadas em /app/forum/moderacao (750)
[OK] Permissões aplicadas em /app/forum/topicos (755)
[INFO] Verificação final:
total 44
drwxr-xr-x 10 root      root      4096 Jun  7 12:01 .
drwxr-xr-x  1 root      root      4096 Jun  7 12:01 ..
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 anexos
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 backups
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 categorias
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 logs
drwxr-xr-x  2 moderador_user forum_ops 4096 Jun  7 12:01 moderacao
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 publicacao
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 topicos
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 usuarios
/etc/group:forum_ops:x:1000:
/etc/passwd:moderador_user:x:999:1000::/home/moderador_user:/bin/false
total 44
drwxr-xr-x 10 root      root      4096 Jun  7 12:01 .
drwxr-xr-x  1 root      root      4096 Jun  7 12:01 ..
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 anexos
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 backups
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 categorias
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 logs
drwxr-xr-x  2 moderador_user forum_ops 4096 Jun  7 12:01 moderacao
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 publicacao
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 topicos
drwxr-xr-x  2 root      root      4096 Jun  7 12:01 usuarios
root@ed78f70bba30:/app#
```

Relatório final gerado -

```

root@ed78f70bba30:/app# bash /app/scripts/09_relatorio.sh
cat /app/logs/relatorio_execucao.txt
=====
RELATÓRIO OPERACIONAL - FÓRUM ONLINE
Aluno: Rafael Zink
Instituição: Unidavi
Data: 2026-06-07 12:07:29
=====

--- ESPAÇO EM DISCO ---
Filesystem      Size  Used Avail Use% Mounted on
overlay          1007G  2.6G  954G   1% /
tmpfs            64M    0   64M   0% /dev
shm              64M    0   64M   0% /dev/shm
C:\              477G  430G   48G  91% /app/logs
/dev/sdd         1007G  2.6G  954G   1% /etc/hosts
tmpfs            3.7G    0   3.7G   0% /proc/acpi
tmpfs            3.7G    0   3.7G   0% /proc/scsi
tmpfs            3.7G    0   3.7G   0% /sys/firmware

--- USO DOS DIRETÓRIOS DO FÓRUM ---
4.0K  /app/forum/anexos
4.0K  /app/forum/backups
4.0K  /app/forum/categorias
8.0K  /app/forum/logs
4.0K  /app/forum/moderacao
4.0K  /app/forum/publicacao
4.0K  /app/forum/topicos
4.0K  /app/forum/usuarios

--- STATUS DO APACHE ---
* apache2 is running

--- ÚLTIMOS BACKUPS ---
total 0
-rw-r--r-- 1 root root 396 Jun  7 12:04 backup_forum_2026-06-07_12-04.tar.gz

--- ÚLTIMOS LOGS ---
total 16K
-rw-r--r-- 1 root root 1.1K Jun  7 12:07 relatorio_execucao.txt
-rw-r--r-- 1 root root  928 Jun  7 12:06 usuarios.log
-rw-r--r-- 1 root root  402 Jun  7 12:04 deploy.log
-rw-r--r-- 1 root root  250 Jun  7 12:04 backup.log
-rw-r--r-- 1 root root  598 Jun  7 12:01 estrutura.log
-rw-r--r-- 1 root root  451 Jun  7 12:01 apache.log
-rw-r--r-- 1 root root 3.9K Jun  7 12:00 update.log

--- ARQUIVOS PUBLICADOS NO APACHE ---
total 20
drwxr-xr-x 1 root root 4096 Jun  7 12:04 .

```

Imagem publicada no DockerHub -

```

root@ed78f70bba30:/app# exit
exit

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug trabalho03-linux
  Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell> docker login
Authenticating with existing credentials... [Username: rafaelzink]

Info → To Login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell> docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
trabalho03-cloud-shell-forum-linux:latest  a1018796087f                    396MB        110MB        U
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell>

```

```
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell> docker tag trabalho03-cloud-shell-forum-linux rafaelzink/trabalho03-forum:latest
PS C:\Users\Rafael\Downloads\trabalho03-cloud-shell> docker push rafaelzink/trabalho03-forum:latest
The push refers to repository [docker.io/rafaelzink/trabalho03-forum]
7055f62b2aee: Pushed
6721fcfc9aac: Pushed
a605c4c911f6: Pushed
4d603208cc60: Pushed
40d16f30db40: Pushed
426fb101a3aa: Pushed
4f4fb700ef54: Pushed
█
```

Menu de execução -

```
=====
Criado por: Rafael Zink
Instituição: Unidavi - 7ª Fase SI
Tema: Infraestrutura para Fórum Online
=====
MENU DEVOPS CLOUD - FÓRUM
=====
1 - Atualizar sistema
2 - Instalar Apache
3 - Criar estrutura do projeto
4 - Realizar backup
5 - Fazer deploy
6 - Ver processos
7 - Monitorar sistema
8 - Configurar usuários e permissões
9 - Gerar relatório
0 - Sair
=====
Escolha uma opção: █
```